

Physical modifications important for consolidating memory are particularly apparent in the hippocampus, especially a pathway called the Schaffer Collateral, which contains CA1 (*cornu Ammonis*) neurons that have been implicated in learning and memory. To study learning and memory in live mice, scientists use a stimulating electrode to generate brief electrical pulses on the dendrites of CA1 neurons in the Schaffer Collateral pathway and use a recording electrode to measure the excitatory postsynaptic potentials (EPSPs) in the CA1 postsynaptic neurons.

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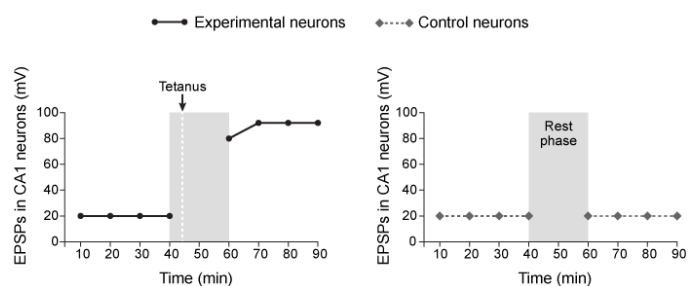


Figure 1 CA1 neuron response after a brief tetanus (experimental) or no tetanus (control)

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Following 30 days in their respective environments, all mice were tested each day for one week in the Morris Water Maze, a test that measures spatial learning and memory consolidation. Results showed that mice in

the experimental condition demonstrated greater improvement on Morris Water Maze performance compared to controls. When hippocampi were examined, the results revealed that mice in the experimental condition had a greater number of dye-labeled cells than did the controls.

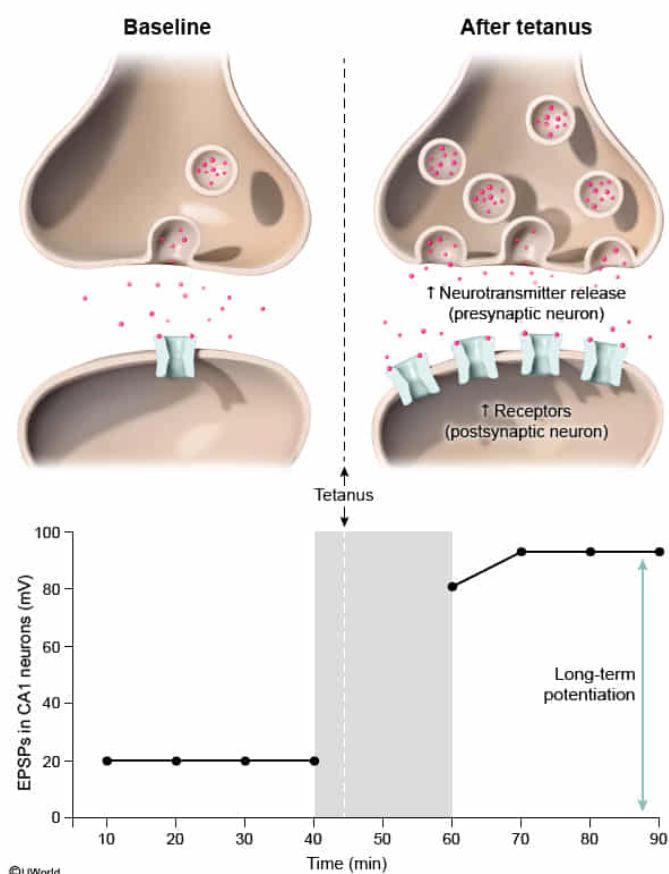
The phenomenon induced by high-frequency stimulation (Figure 1) is referred to as:

- ☐ A. sensory adaptation. [10%]
- ☐ B. state-dependent memory. [2%]
- ✓ ☒ C. long-term potentiation. [63%]
- ☐ D. tonic stimulus response. [23%]

Correct

63% answered correctly

Explanation:



Long-term potentiation (LTP) describes an enduring increase in synaptic transmission of neurons, which is the neural foundation for **learning** and **memory** consolidation. When neurons are repeatedly stimulated, they

demonstrate an increased firing rate, known as LTP. The increase in magnitude of excitatory postsynaptic potentials (EPSPs) in the postsynaptic neurons in the experimental condition (Figure 1) reflects LTP.

LTP can also occur when two (or more) neurons repeatedly fire simultaneously. For example, if a neuron repeatedly receives simultaneous input from two sources, neuron A (visual input of flower) and neuron B (smell of flower), the combined inputs may become associated. If this occurs, either input A or input B will activate the postsynaptic neuron, but others, like input C (smell of an onion), will not. This is the neural foundation for how we learn **associations**.

LTP can occur by two mechanisms: An increase in the release of neurotransmitter by the presynaptic neuron and an increase in the number of receptors in the postsynaptic neuron.

(Choice A) Sensory adaptation (or neural adaptation) is a decreased responsiveness of a sensory neuron over time in response to a constant stimulus. In other words, when a stimulus doesn't change, the firing rate of the neuron detecting that stimulus declines. Sensory adaptation occurs in the peripheral (not central) nervous system and would produce a graph with a decrease (not increase) in EPSPs.

(Choice B) State-dependent memory is a phenomenon whereby memory retrieval is most efficient when an individual's internal state at the time of retrieval is the same as when the memory was encoded. For example, memories encoded while an individual is intoxicated are less easily recalled when sober than when intoxicated at a later time.

(Choice D) Tonic receptors are sensory receptors that continue to produce action potentials throughout the duration of a stimulus. Although a tonic stimulus response would produce graph results similar to those of the experimental neurons in Figure 1, tonic receptors are limited to the peripheral nervous system and are not found in the brain.

Educational objective:

Long-term potentiation (LTP) describes the neural changes responsible for learning, memory, and associations. LTP occurs when a neuron's firing rate increases after repeated stimulation or simultaneous stimulation by multiple inputs (association). LTP results from an increase in neurotransmitter

release and postsynaptic receptors.

Time spent:0 sec

Subject:
Behavioral Sciences

QID:400631

System:
Learning, Memory, and Cognition

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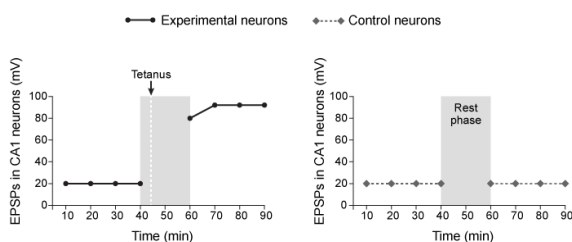


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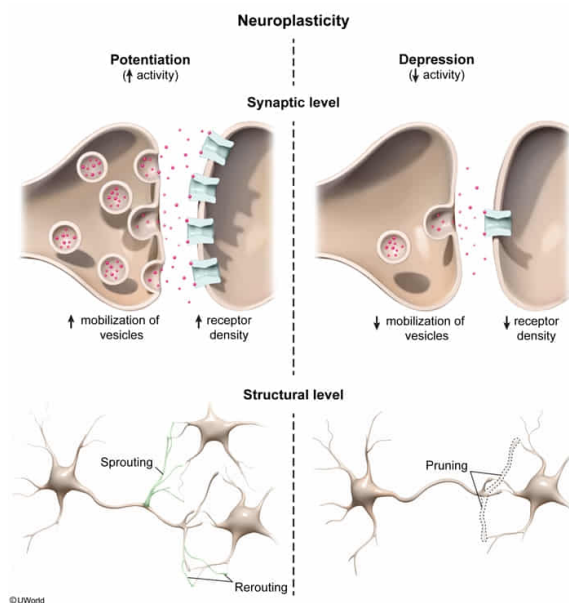
Which of the following conclusions regarding the neuroplasticity of CA1 neurons in the experimental condition is best supported?

- ☐ A. Presynaptic neurons demonstrate synaptic plasticity [7%]
- ☐ B. Presynaptic neurons demonstrate structural plasticity [1%]
- ☒ C. Postsynaptic neurons demonstrate synaptic plasticity [80%]
- ☐ D. Postsynaptic neurons demonstrate structural plasticity [10%]

Correct

79% answered correctly

Explanation:



Neuroplasticity (or neural plasticity) refers to **lasting changes in the brain** that occur when interactions with the environment alter neurons and/or pathways. Neuroplasticity refers to both increases in neuronal connections, known as **potentiation**, and decreases in neuronal responses, known as **depression**. Neuroplasticity is highest during early development but continues throughout our lifetime.

Neuroplasticity can occur on the synaptic or structural level. Synaptic plasticity results from changes in the firing rate of the presynaptic neuron, which alters the amount of neurotransmitter released into the synaptic cleft and the number of receptors on the postsynaptic target. Synaptic changes are associated with both immediate and more delayed potentiation or depression. At the structural level, sprouting (increased connections between neurons), rerouting (new connections between neurons), and pruning (decreased connections between neurons) contribute to structural plasticity. Structural plasticity does not happen quickly, so it is not responsible for immediate changes (**Choice D**).

Study 1 demonstrated long-term potentiation (LTP) within a relatively immediate time frame (90 minutes). Therefore, the neuroplasticity demonstrated by the postsynaptic neurons was due to synaptic plasticity.

(Choices A and B) Only *postsynaptic* CA1 neurons were tested. Although presynaptic neurons can demonstrate synaptic plasticity resulting in immediate LTP, presynaptic CA1 neurons were not assessed in Study 1.

Educational objective:

Neuroplasticity refers to physical changes that occur in the brain throughout a lifetime, including both increases (potentiation) and decreases (depression) in neuronal

responses/connections. Neuroplasticity occurs through physical modifications at the synapses (immediate and delayed plasticity) and the structural level (delayed plasticity only).

Time spent:0 sec

Subject:
Behavioral Sciences

QID:400632

System:
Learning, Memory, and Cognition

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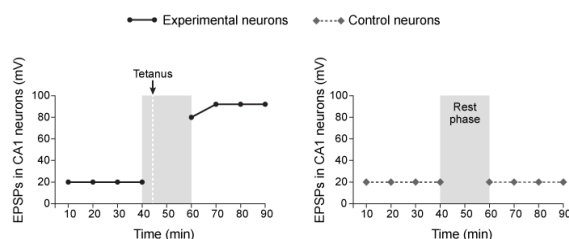


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The results of Study 2 suggest that:

- I. enhanced environments increase cell proliferation in the hippocampus.
- II. cell proliferation increases performance on the Morris Water Maze.
- III. swimming in the Morris Water Maze increases cell proliferation in the hippocampus.

✔ ☒ A. I and II only [81%]

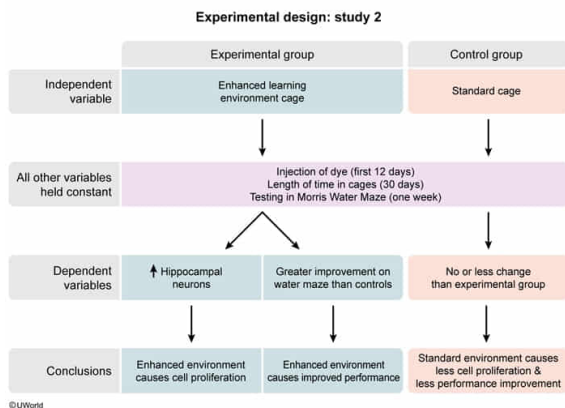
☐ B. I and III only [7%]

- ☐ C. II and III only [2%]
- ☐ D. I, II, and III [9%]

Correct

81% answered correctly

Explanation:



Experimental designs allow for **causal** (or cause-and-effect) conclusions. The manipulation of one variable (the independent variable) while carefully controlling for all other variables allows the researcher to conclude that changes in an outcome of interest (the dependent variable) are *caused by* the independent variable.

In Study 2, the environment in which the mice were housed ("standard" for the control group vs. "enhanced" for the experimental group) was the independent variable, and there were two dependent variables: Performance in the Morris Water Maze and proliferation of cells in the hippocampus.

(Number I) The mice in the experimental condition had a greater number of dye-labeled cells than did the controls, which demonstrates an increase in the number of neurons (neurogenesis) in the hippocampus. Therefore, it can be concluded that an enhanced environment increases cell proliferation in the hippocampus.

(Number II) The mice in the experimental condition performed better in the Morris Water Maze, a test of spatial learning and memory consolidation. Cell proliferation in the hippocampus, a region important for spatial learning and memory consolidation, was demonstrated by an increase in dye-labeled cells; therefore, it is also reasonable to conclude that cell proliferation increases performance on the water maze.

(Number III) Cell proliferation (neurogenesis) and increased performance on the Morris Water Maze were evident only in the experimental condition *as a result of the enhanced environment*. If cell proliferation had occurred as a result of the Morris Water Maze, which occurred last, there should have been no significant differences between the experimental and control groups.

Educational objective:

Causal conclusions can be drawn from well-designed experiments that carefully control for all variables. Manipulation of the independent variable allows the researcher to conclude

that changes to the dependent variable(s) are *caused by* the independent variable.

Time spent:0 sec

Subject:
Behavioral Sciences

QID:400633

System:
Learning, Memory, and Cognition

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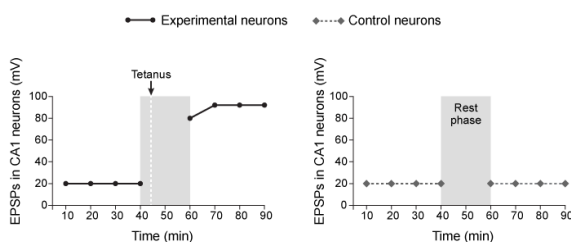


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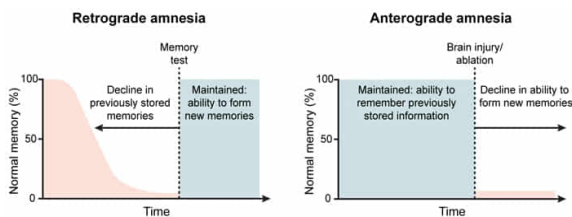
If Study 2 were repeated with a third group of mice with ablated hippocampi (ablation condition), this condition would be most expected to:

- ☐ A. perform as well as the controls on the Morris Water Maze. [2%]
- ☒ B. perform worse than the controls on the Morris Water Maze. [91%]
- ☐ C. have a greater number of dye-labeled cells than the controls have. [2%]
- ☐ D. have an equivalent number of dye-labeled cells as the controls. [2%]

Correct

92% answered correctly

Explanation:



Ablation (the removal or destruction of tissue) is a technique used to study the brain by measuring the deficits that result from ablation of specific brain areas. Bilateral ablation of the **hippocampus**, a region that plays an important role in the consolidation of short-term memory into long-term memory, results in severe **anterograde amnesia**, which is the **inability to form new memories**.

Anterograde amnesia was induced accidentally in the famous case of H.M., a man whose hippocampi were removed bilaterally to eliminate debilitating seizures. As a result, he could no longer form new memories, and to some extent he also experienced **retrograde amnesia**, which is **memory loss** for events **previously encoded**.

If a third group of mice with hippocampal ablations were added to the experiment and tested using the Morris Water Maze, they would be expected to perform worse than either the experimental group or the control group as the Morris Water Maze requires the ability to consolidate memory.

(Choice A) The mice in the ablation condition are unable to consolidate memory at all, which would result in worse performance on the water maze than seen with control mice with intact hippocampi.

(Choices C and D) The injected dye was incorporated into newly synthesized DNA in the hippocampus, providing a biomarker for neuronal growth in this area due to learning. Ablation of the hippocampus would result in no dye-labeled cells in this area.

Educational objective:

Ablation of the hippocampus produces anterograde amnesia, the inability to form new memories. However, bilateral hippocampal ablation would also result in some degree of retrograde amnesia, the inability to recall memories that were previously encoded. The hippocampus controls memory consolidation.

Time spent: 0 sec

Subject:
Behavioral Sciences

QID: 400634

System:
Learning, Memory, and Cognition

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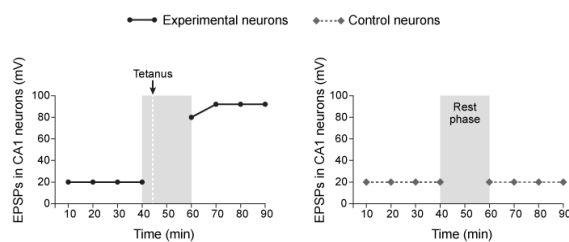


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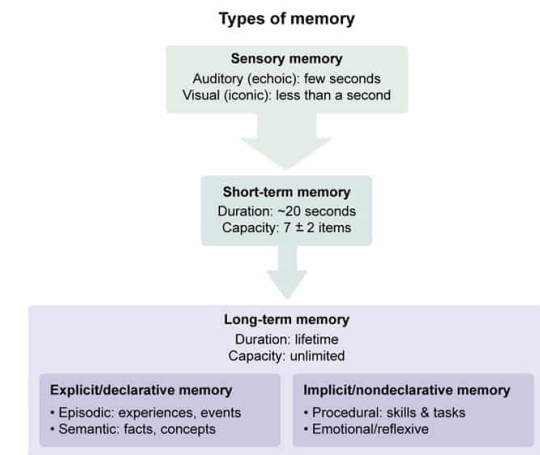
In Study 2, what type of human memory is most analogous to the type of memory assessed in mice?

- ☐ A. Sensory [12%]
- ☐ B. Episodic [23%]
- ☐ C. Semantic [14%]
- ✓ ☒ D. Implicit [49%]

Correct

49% answered correctly

Explanation:



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Research suggests that humans have two major, long-term memory systems, referred to as explicit and implicit. **Implicit memory** does not require conscious thought. A type of implicit memory, **procedural memory** involves learning **motor skills**, such as riding a bicycle, playing an instrument, or tying shoes. In Study 2, the water maze is a spatial and motor task requiring information that is most analogous to what might be stored in implicit memory.

Explicit memory can be consciously or intentionally recalled. Explicit memory is often referred to as declarative memory, which is memory for facts and events. Declarative memory includes episodic memory, which is the memory for personal experiences (eg, your first day of college), and semantic memory, which includes fact-based knowledge (eg, name of the first US president) and language (eg, "their" and "there" are not synonymous).

(Choice A) Sensory memory is the very brief temporary storage of sensory information (eg, sights, sounds, smells). Much information comes into the sensory memory store, but anything that does not move on to the next system, short-term memory, is almost immediately lost. Although sensory memory is required to complete spatial/motor tasks like the Morris Water Maze, it is not what is being assessed in the second study.

(Choices B and C) Episodic and semantic memory are forms of explicit memory, which is not the type of memory required to solve a spatial/motor task such as the Morris Water Maze.

Educational objective:

Procedural memory, a type of implicit memory, is long-term memory for learned motor skills and is not consciously accessible. Explicit memory, or declarative memory, is long-term memory for facts (semantic) or personal events (episodic) and is consciously accessible.

Time spent: 0 sec

Subject:

Behavioral Sciences

QID: 400635

System:

Learning, Memory, and Cognition

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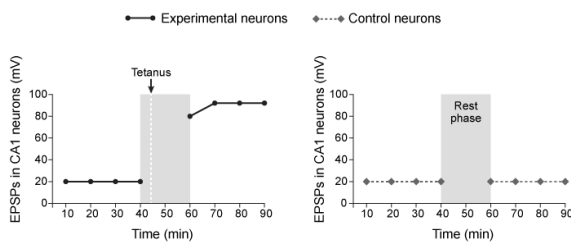


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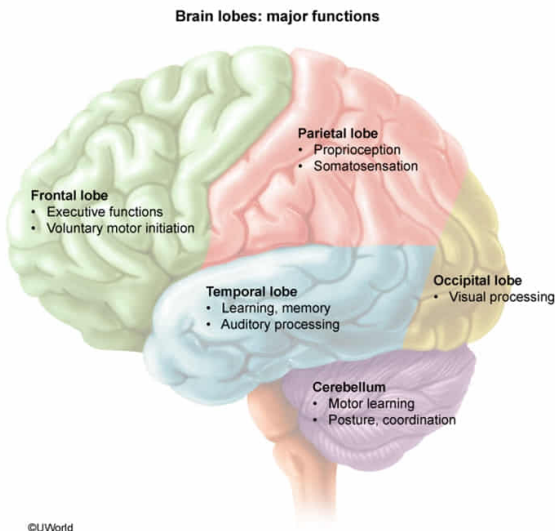
Which of the following correctly pairs a brain area vital for learning in the Morris Water Maze with its appropriate function?

- ✔ ☒ A. Cerebellum; motor learning [30%]
☐ B. Frontal lobe; motor coordination [22%]
☐ C. Occipital lobe; spatial processing [38%]
☐ D. Hypothalamus; sensory relay [9%]

Correct

30% answered correctly

Explanation:



The **cerebellum** is critical for **motor learning**, which occurs whenever an organism repeatedly practices a motor task, such as swimming, driving, or speaking. Motor learning is the foundation for procedural memory, the acquisition of which relies heavily on the cerebellum. Because the mice were tested in the Morris Water Maze each day for one week, the cerebellum would be involved in motor learning of this skill.

The cerebellum is also responsible for integrating information from the visual, vestibular, and somatosensory systems to maintain **balance** and **posture**, **coordinate** complex movements requiring multiple muscles, and control **precision** of fine motor movements. The cerebellum also receives feedback from the body regarding the position of body parts in space and relative to each other (**proprioception**).

(Choice B) The frontal lobe of the brain is responsible for many functions, including the active manipulation of information involved in reasoning and problem solving (known as working memory) and the initiation of voluntary motor movements. The *cerebellum* (not the frontal lobe) ensures motor coordination by balancing excitatory and inhibitory neural impulses so movement is smooth and precise rather than jerky.

(Choice C) The occipital lobe is primarily responsible for processing visual information. The *parietal lobe* (not the occipital lobe) processes spatial information (proprioception) and is responsible for cognitive mapping, our ability to represent mentally the spatial relationships concerning things in our environment.

(Choice D) The hypothalamus is important for many metabolic processes that maintain homeostasis. The *thalamus* (not the hypothalamus) is responsible for relaying sensory information to the cerebral cortex.

Educational objective:

The cerebellum is central to motor learning, maintaining balance and posture, coordinating

complex movements, and controlling precision of fine motor movements. The cerebellum does not initiate voluntary movement, which occurs in the primary motor cortex of the frontal lobe; however, the cerebellum is critical for most other aspects of motor planning and execution.

Time spent:0 sec

Subject:
Behavioral Sciences

QID:400636

System:
Learning, Memory, and Cognition